Grenoble, France

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Zahra KANANI

Artificial Intelligence for Healthcare | Biomedical Engineering

Biomedical Engineer with expertise in Artificial Intelligence for Healthcare

Bridging engineering and AI to advance medical research and patient care. Experienced in applying machine learning to biomedical data, delivering practical and innovative healthcare solutions.

Technical Skills

Artificial Intelligence & Data Science

- **Programming:** Python, R
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, spaCy,
- **Methods:** CNNs, Semi-supervised Learning, Transfer Learning
- **Visualization:** Matplotlib, TensorBoard

Biomedical Engineering & Regulation

- **Modeling & Simulation:** Abaqus, SolidWorks, CATIA
- **Biomedical Applications:** Orthopedic implant, Cardiovascular devices, Tissue modeling
- **Regulatory Standards:** ISO 13485, ISO 14971, ISO 10993, EU MDR

Awards and honors

- Recipient of a scholarship from the Multidisciplinary Institute for Artificial Intelligence (MIAI).
- Earlier academic honors: direct M.Sc. admission (2013) and Talented Students Center membership (2009–2013).

Languages

- English (Fluent)
- French (intermediate)

Artificial Intelligence Experience

Master 2 AI for One Health (AI4OneHealth)

Grenoble Alpes University, Grenoble, France (2024-2025) | Supervisor: Pascal Mossuz

- **Relevant Courses:** Ethical and societal aspects of AI, Machine Learning and Deep Learning for Health, Medical Text Mining, Application of AI for Healthcare

Master 2 Internship

Laboratoire d'Informatique de Grenoble (LIG), Université Grenoble Alpes (May–Sep 2025) | Supervisor: Prof. Masih Reza Amini and Dr.Aude Sportisse

- Compared FixMatch and AdaMatch under class imbalance on CIFAR-10.
- Proposed an AdaMatch–FlexMatch hybrid method that improved minority-class accuracy while preserving majority-class performance.

Selected AI Projects

- Biomedical Text Classification: Compared TF-IDF and PubMedBERT.
- Information Retrieval: Built a Python-based search engine with PyTerrier.
- COVID-19 Abstract Analysis: Processed and analyzed abstracts using spaCy and MetaMap.
- Image Classification with CNN: Applied TensorFlow/Keras for digit and satellite image recognition.

Certification

TensorFlow 2, Imperial College London, Coursera (Link)

Biomedical Engineering and Regulatory Experience

Regulatory Specialist

Food and Drug Administration of Iran , Tehran, Iran, 2017-2022

- Evaluated and regulated medical devices, with a focus on orthopedic and cardiovascular devices.

Research Assistant

Orthopedic and Dental Biomechanics Lab (ODBL), AmirKabir University of Technology, Iran, 2015-2017

- Conducted experimental and numerical analyses for orthopedic implant studies.

Bachelor & Master in Biomedical Engineering (Bio-Mechanics)

AmirKabir University of Technology, Tehran, Iran (2009-2015)

- **Master Thesis:** Optimization of large intestine anastomosis using Finite Element Method.
- **Bachelor Thesis:** Effect of Poly-L-lactic material on bone–plate fixation stability.

Publication

Biocybernetics and Biomedical Engineering (2019). "Investigating the effects of suturing parameters on leakage from intestinal anastomosis: finite element analyses." DOI

Certification

ISO 13485:2016 – Medical Devices QMS (BSI Training Academy, 2021)